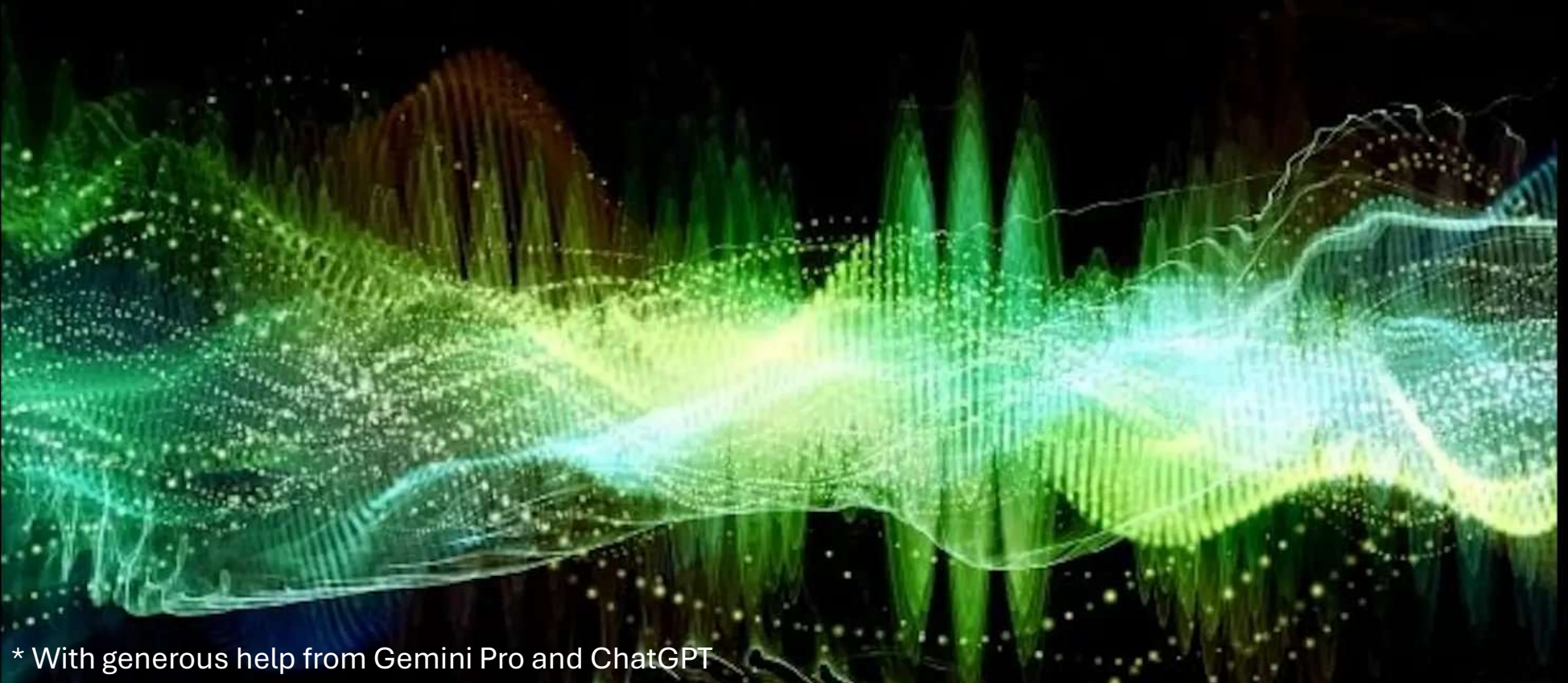


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 1, 8-20-26, 8-25-26

Peter Mathys*



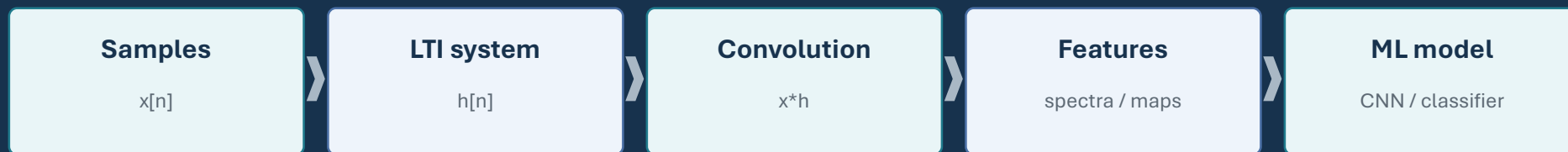
* With generous help from Gemini Pro and ChatGPT

DSP + ML

Signals, LTI Systems, and Convolution

Week 1 • Lecture 1 • 75 minutes

Classical DSP foundations → modern signal representations → machine learning



Today's learning objectives

00–15 MIN

By the end, students should be able to reason—not just compute.

- **Represent discrete-time signals using impulses, steps, and complex exponentials.**
- Test systems for linearity, time invariance, causality, and BIBO stability.
- Explain why the impulse response completely characterizes an LTI system.
- Compute finite-length convolutions graphically and numerically.
- Connect convolution to filtering and to the operation used by neural networks.

Course habit

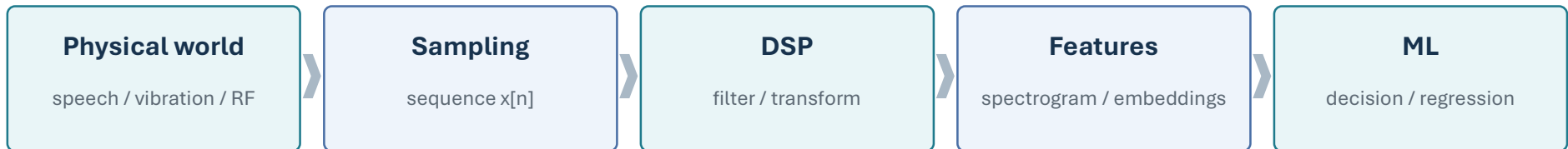
For each new equation, ask: What does this let us predict about a real signal?

Why start with LTI systems?

Because many useful DSP operations become simple, interpretable building blocks.

→ enables us to use mathematically closed form solutions

00-15 MIN



Classical DSP gives us...

- Structure: linearity and invariance
- Closed-form analysis
- Stability and causality tests
- Frequency-selective filtering

Amplifiers LTI
Equalizers LTI
Compressor not LTI
Limiters not LTI

ML later needs...

- Well-conditioned inputs
- Meaningful local structure
- Efficient operators
- Interpretability and diagnostics

Discrete time: index n is not “time in seconds”

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n labels samples; physical time appears only after a sampling rate is specified.

$$t_n = nT_s = \frac{n}{f_s}$$

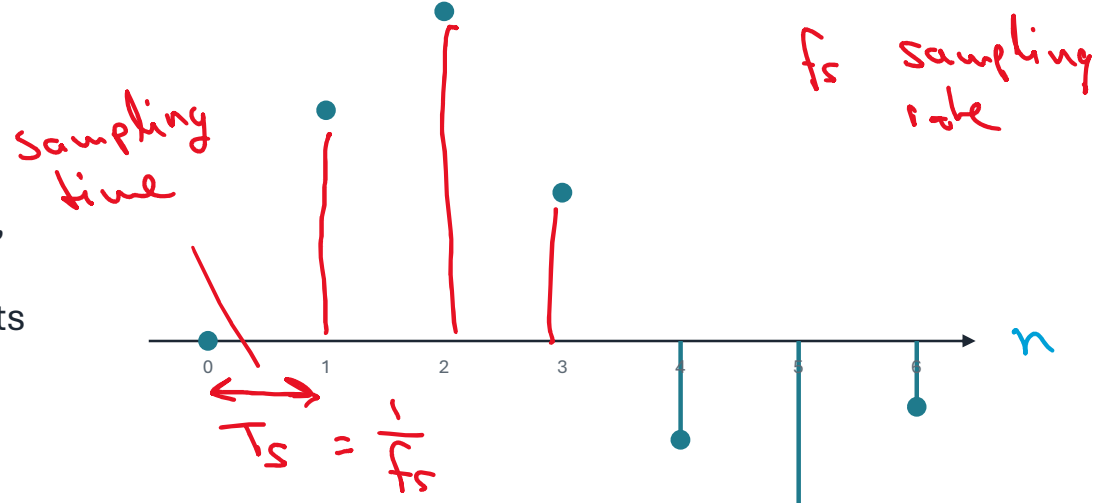
- $x[n]$ is defined only at integer n.
- A shift $x[n-n_0]$ changes the sample index, not the amplitude.
- Normalized angular frequency ω has units rad/sample.

$x[n] = 0, 2, 3, 1$
 ↑ ↑ ↑
 n=0 n=1 n=2

n=10 at which time does it happen?

for any n $t_n = nT_s = \frac{n}{f_s}$

Example sequence $x[n]$



Audio connection

At 48 kHz, one sample is 20.83 μ s. A 2400-sample delay is 50 ms.

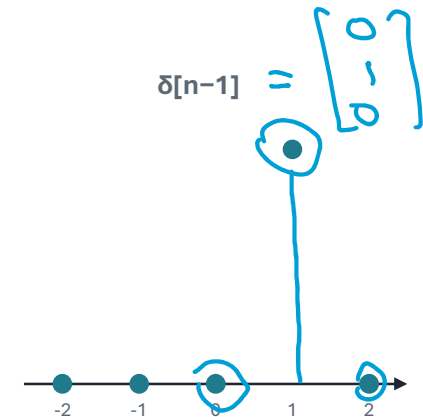
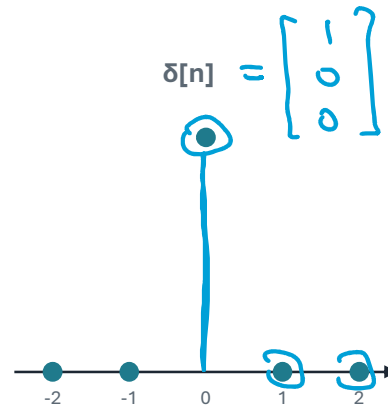
$$\frac{2400}{48000} = \frac{1}{20} = 0.05$$

The unit sample $\delta[n]$ is a coordinate basis

15-35 MIN

It isolates one sample at one index.

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

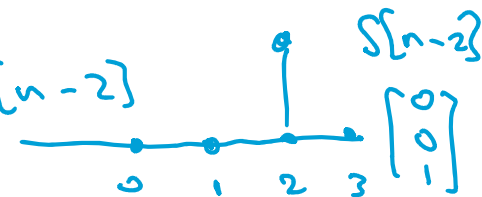


expanding $\underline{x}$ using multidimensional basis vectors $\delta[n-k]$

$$\underline{x}[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

$\delta[n]$ and $\delta[n-1], \delta[n-2], \dots$ are basis vectors

$$\underline{x} = x_0 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = x_0 \delta[n] + x_1 \delta[n-1] + x_2 \delta[n-2]$$



Interpretation

A sequence is a weighted sum of shifted impulses—the discrete-time analog of expanding a vector in basis vectors.

$$x[n] = \begin{cases} 2 & n=0 \\ -1 & n=1 \\ 3 & n=3 \\ 0 & \text{otherwise} \end{cases}$$

$$= 2\delta[n] - \delta[n-1] + 3\delta[n-3] + 0\delta[n-2]$$

why is there no $\delta[n-2]$

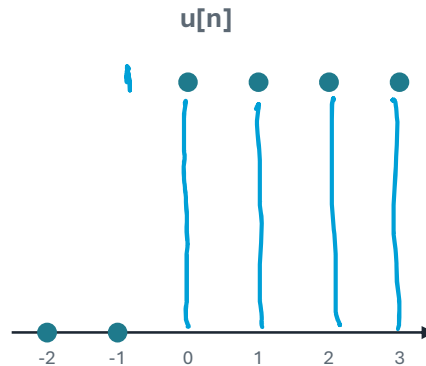
$$x[n] = \sum_k x[k] \cdot \delta[n-k]$$

The unit step $u[n]$ and useful identities

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Steps describe turn-on behavior; impulses describe changes.

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



$$\delta[n] = u[n] - u[n-1]$$

$$u[n] = \sum_{k=-\infty}^n \delta[k]$$

Quick check

What is $u[n+2]$? Where does it turn on?

$$n = -2 \quad n \geq -2$$

$$u[n-4] \quad \text{where does it turn on?}$$

$n = 4 \quad n \geq 4$

$$u[-n+3] \quad \text{where does it turn on?}$$

off

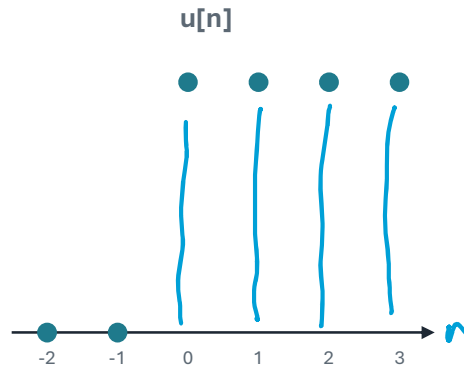
$$-n+3 \geq 0 \quad n \leq 3$$

The unit step $u[n]$ and useful identities

15-35 MIN

Steps describe turn-on behavior; impulses describe changes.

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



$$\delta[n] = u[n] - u[n-1]$$

$$u[n] = \sum_{k=-\infty}^n \delta[k]$$

$= \sum_{k=0}^{\infty} \delta[n-k]$

Quick check

What is $u[n+2]$? Where does it turn on?

Answer

At $n = -2$. A “+2” inside the brackets shifts the sequence left by 2.

$$u[n] = \delta[n] + \delta[n-1] + \delta[n-2] + \dots$$
$$= \sum_{k=0}^{\infty} \delta[n-k]$$

Complex exponentials are the “natural modes” of LTI systems

15-35 MIN

Today: recognize them. Lecture 2: prove the eigenfunction property.

$$x[n] = A\alpha^n e^{j\omega_0 n}$$

$$e^{j\omega_0 n} = \cos(\omega_0 n) + j\sin(\omega_0 n)$$

Euler's identity

What each factor does

- A sets scale / phase
- α^n gives growth or decay
- $e^{j\omega_0 n}$ gives rotation / oscillation

$|\alpha| < 1$ decaying function

what happens if $\alpha = 1$?

Why DSP cares

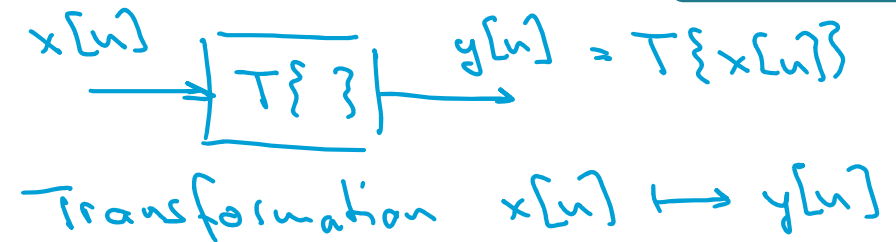
- Sinusoids probe frequency response
- Exponentials diagonalize convolution
- Spectra decompose signals into modes

if $x[n] = A\alpha^n e^{j\omega_0 n}$
goes into LTI system
output is at same freq ω_0
with amplitude/phase change

What is a discrete-time system?

A rule that maps an input sequence to an output sequence.

$$y[n] = T\{x[n]\}$$



Examples

- Gain: $y[n] = 2x[n]$
- Delay: $y[n] = x[n-3]$
- Moving average: $y[n] = (x[n] + x[n-1])/2$

Nonlinear / time-varying examples

- Squaring: $y[n] = x^2[n]$
- Compressor / limiter
- Gain that changes with n

Question to ask

Which properties let us predict behavior without simulating every possible input?

linearity, time-invariance, causality, stability

Linearity = superposition

Scale inputs → scale outputs. Add inputs → add outputs.

$$T\{ax_1[n] + bx_2[n]\} = aT\{x_1[n]\} + bT\{x_2[n]\}$$

Linear

- $y[n] = 3x[n]$
- $y[n] = x[n] + 0.5x[n-1]$
- Any convolution with fixed $h[n]$

Linearity: Mental decomposition:

Scaling: $x[n] \mapsto y[n] \Rightarrow ax[n] \mapsto ay[n]$

Addition $x_1[n] \mapsto y_1[n], x_2[n] \mapsto y_2[n]$

$\Rightarrow x_1[n] + x_2[n] \mapsto y_1[n] + y_2[n]$

Superposition

Not linear

- $y[n] = x[n] + 2$
- $y[n] = |x[n]|$
- Dynamic-range compressor

Simple test (not proof)
Set $x[n] = 0$ then
 $y[n]$ must = 0
if linear

Fast test

A linear system must map zero input to zero output.

Time invariance = “the clock does not matter”

Shift the input; the output shifts by exactly the same amount.

$$T\{x[n - n_0]\} = y[n - n_0] \quad \text{when } y[n] = T\{x[n]\}$$

Time invariant

- Fixed delay
- Fixed FIR / IIR filter
- Static gain

Use a two path test
 path 1: compute output, then shift
 path 2: shift input, then compute output

Time varying

- $y[n] = n \cdot x[n]$
- Daily adaptive equalizer
- Any rule using absolute sample index n

$y[n - n_0] = (n - n_0) \cdot x[n - n_0]$ — $\neq$
 $x[n - n_0] \rightarrow n \cdot x[n - n_0]$ — $\neq$
 $\Downarrow$
 not LTI

Audio analogy

An echo with a fixed 100 ms delay is time invariant. An echo whose delay sweeps over time is not.

Causality and BIBO stability

35-55 MIN

Two properties with direct implementation consequences.

Causal system

- $y[n_0]$ depends only on $x[n]$ for $n \leq n_0$
- Can operate in real time
- For LTI: $h[n] = 0$ for $n < 0$

output now cannot depend
on future

For LTI system need $h[n] = 0, n < 0$
for causality

$$\|x[n]\| \leq B_x < \infty \Rightarrow \|y[n]\| \leq B_y < \infty$$

Non causal works for recorded sequences
Audio editor does not need to be causal

BIBO stable system

- Every bounded input gives bounded output
- For LTI: $\sum |h[n]| < \infty$
- Rules out runaway responses

assume input $|x[n]| \leq B_x$

BIBO stable we need

$$|y[n]| = \sum_k |x[k]| |h[n-k]| \leq B_y$$

$B_y < \infty$

Graduate stretch

Be able to prove the LTI stability condition using $|x * h| \leq B_x \sum |h|$.

Property check: classify these audio systems

35-55 MIN

Think → vote → justify.

For each system: linear? time invariant? causal? BIBO stable?

linear time invariant causal BIBO stable

A $y[n] = x[n-4800]$

Fixed 100 ms delay at 48 kHz

✓ ✓ ✓ ✓

B $y[n] = 0.8x[n] + 0.2x[n-1]$

Two-tap smoother

✓ ✓ ✓ ✓

C $y[n] = |x[n]|$

Full-wave rectifier

— ✓ ✓ ✓

D $y[n] = n x[n]$

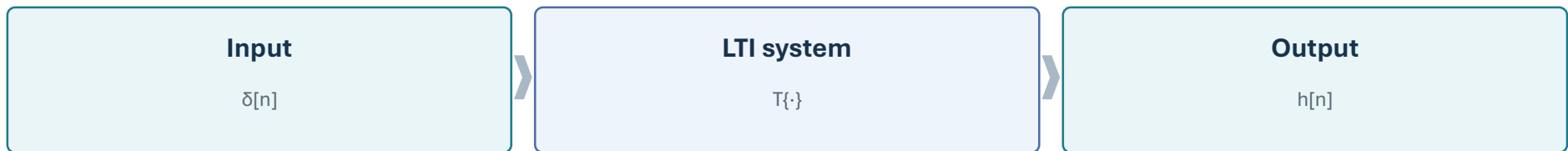
Index-dependent gain

✓ — ✓ —

Impulse response: one experiment characterizes an LTI system

55-75 MIN

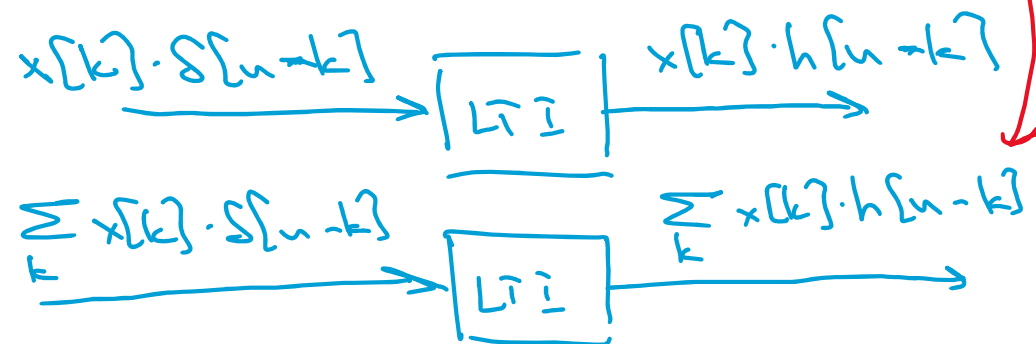
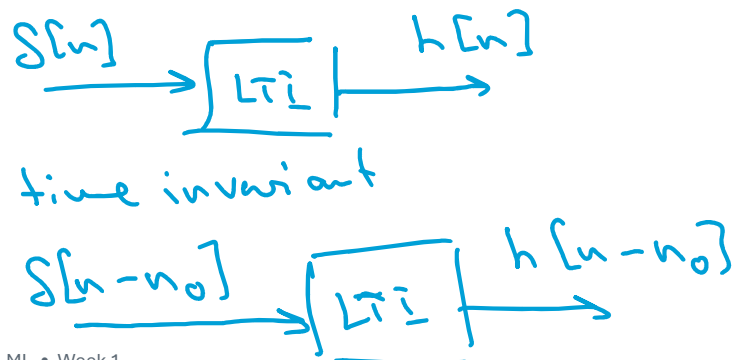
Feed $\delta[n]$; call the output $h[n]$.



$$h[n] = T\{\delta[n]\}$$

we developed convolution!

- Linearity tells us how scaled impulses behave.
- Time invariance tells us how shifted impulses behave.
- Since any $x[n]$ is a sum of scaled, shifted impulses, $h[n]$ tells us the response to any $x[n]$.



Deriving convolution from impulse decomposition

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This derivation is worth understanding line by line.

$$x[n] = \sum_k x[k] \delta[n - k]$$

1. Decompose the input into shifted impulses.

$$T\{x[n]\} = \sum_k x[k] T\{\delta[n - k]\}$$

2. Use linearity.

$$T\{\delta[n - k]\} = h[n - k]$$

3. Use time invariance.

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] \equiv \underline{x[n] * h[n]}$$

Meaning

Each input sample launches a shifted copy of $h[n]$; the output is their sum.

$$\begin{array}{ccc} x[n] = \sum_k x[k] \cdot \delta[n-k] & & \\ \downarrow & \swarrow \leftarrow T\{\cdot\} \text{ of LTI system} & \\ y[n] = \sum_k x[k] h[n-k] & & \end{array}$$

Convolution mechanics: flip → shift → multiply → add

55-75 MIN

For finite sequences, the overlap changes as n moves.



$$y[n] = \sum_k x[k]h[n - k]$$

Graphical view

- Track overlap as n changes
- Good for intuition
- Best with short sequences

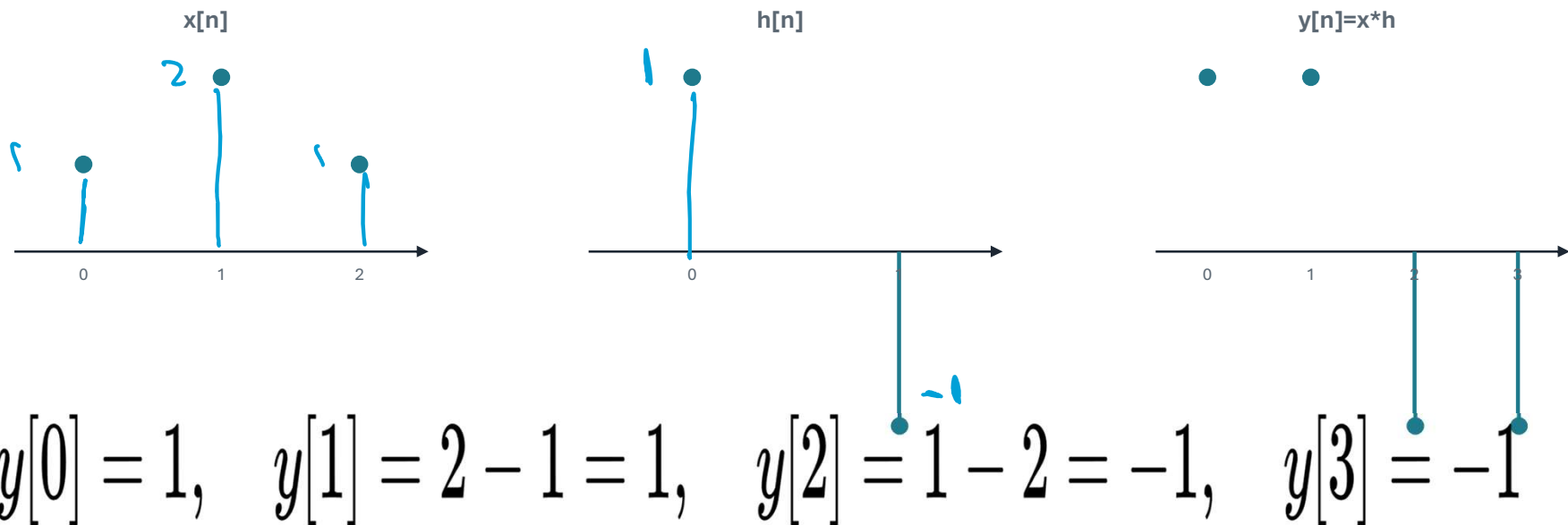
Algebraic / code view

- Use nonzero-support limits
- Good for longer signals
- `np.convolve` / `scipy.signal`

Worked convolution: $x[n]=[1,2,1]$, $h[n]=[1,-1]$

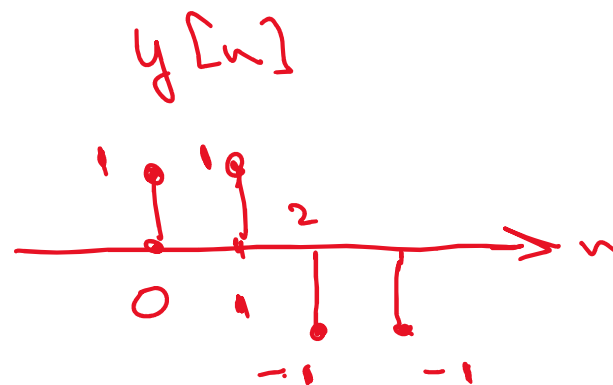
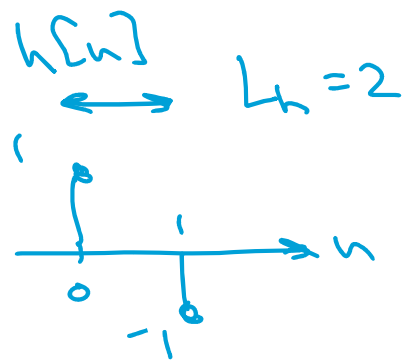
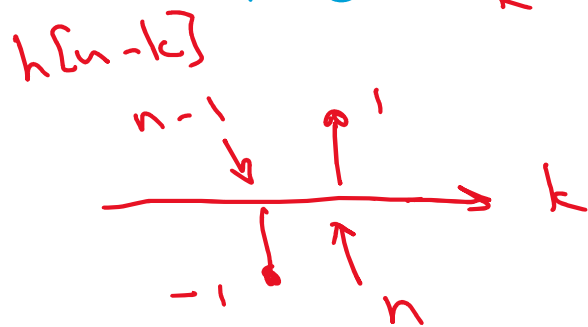
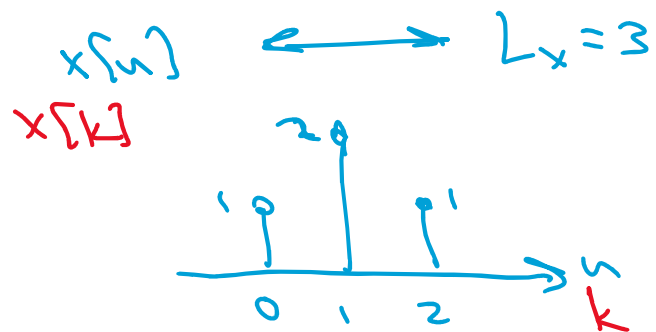
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Assume both start at $n=0$.



Sanity checks

Output length = $3+2-1 = 4$. $\text{Sum}(y) = \text{sum}(x)\text{sum}(h) = 0$.



Support: $L_x + L_h - 1 = 4 = L_y$

$$\sum_n y[n] = \left(\sum_n x[n] \right) \cdot \left(\sum_n h[n] \right) = 0$$

$= 4 \qquad = 0$

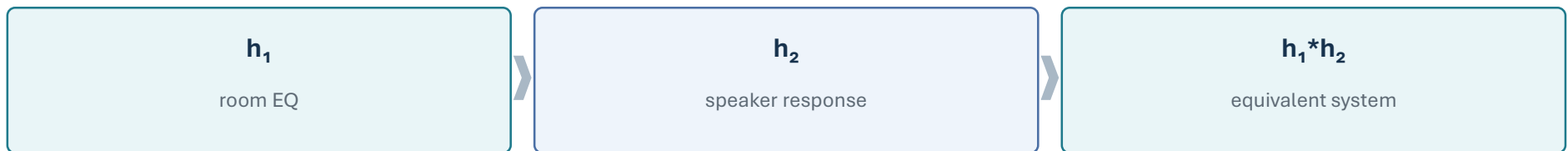
Convolution has algebraic structure

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Useful facts that save work and explain cascades.

$$x * h = h * x \quad x * (h_1 * h_2) = (x * h_1) * h_2 \quad x * (h_1 + h_2) = x * h_1 + x * h_2$$

commutative associative distributive

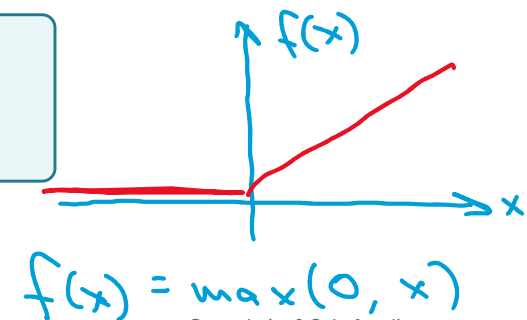


Why ML cares

A stack of linear convolution layers with no nonlinearity collapses to one convolution.

CNNs use ReLU between convolution layers

ReLU Rectified Linear Unit



Oppenheim & Schaffer alignment

Python demo: convolution in one line

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Then verify the indexing and meaning.

Python

```
import numpy as np

x = np.array([1., 2., 1.])
h = np.array([1., -1.])
y = np.convolve(x, h, mode='full')
print(y)  # [ 1.  1. -1. -1.]
```

SciPy

```
# For long signals, scipy.signal.fftconvolve
from scipy.signal import fftconvolve
y_fast = fftconvolve(x, h, mode='full')
```

Do not skip the math

A library call gives numbers. Convolution theory tells you support, delay, stability, frequency behavior, and what to debug.

ML bridge: PyTorch “convolution” is usually cross-correlation

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Same local weighted sum, different kernel orientation.

DSP convolution: $y[n] = \sum_k x[k]h[n-k]$ CNN correlation: $y[n] = \sum_k w[k]x[n+k]$

weights (pointing to $w[k]$) *features* (pointing to $x[n+k]$)

correlation (underlined below the sum)

PyTorch

```
import torch
import torch.nn.functional as F

x = torch.tensor([[[[1., 2., 1.]]]])
w = torch.tensor([[[[1., -1.]]]])
y = F.conv1d(x, w) # correlation convention
```

Key idea

The kernel is learned in ML, but the local linear operation is the same family of operation you are analyzing now.

conv 2d

Concept check

CHECK

No computation should be necessary.

For an LTI system, which statement is always true?

A. If $x[n]$ is bounded, $y[n]$ is bounded.

B. The response to $\delta[n-k]$ is $h[n-k]$.

C. The system must be causal.

D. The impulse response has finite length.

Concept check

CHECK

No computation should be necessary.

For an LTI system, which statement is always true?

A. If $x[n]$ is bounded, $y[n]$ is bounded.

B. The response to $\delta[n-k]$ is $h[n-k]$.

C. The system must be causal.

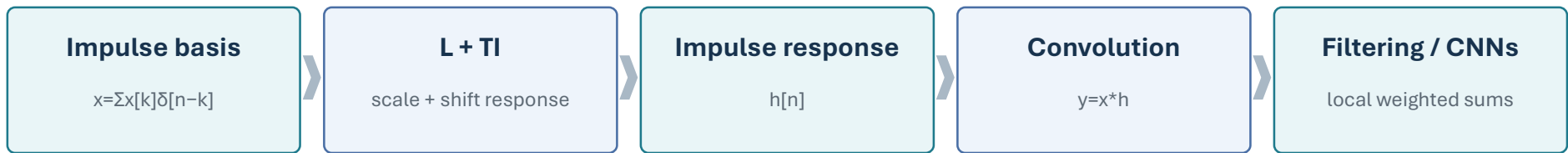
D. The impulse response has finite length.

Correct: time invariance shifts the impulse response; linearity handles scaling. Stability and causality are additional properties, not automatic.

Lecture 1 synthesis

WRAP-UP

The chain of reasoning matters more than memorizing isolated formulas.



- Be able to test system properties with a counterexample or proof.
- Be able to derive convolution from LTI assumptions.
- Be able to compute a short convolution by hand and predict its support.

Next lecture

Complex exponentials turn convolution into multiplication in the frequency domain.

Optional board reserve: index bookkeeping for finite convolution

RESERVE

Use if students need one more example.

$$x[n] \neq 0 \text{ on } [n_x^{\min}, n_x^{\max}], \quad h[n] \neq 0 \text{ on } [n_h^{\min}, n_h^{\max}]$$

$$y[n] \neq 0 \text{ only on } [n_x^{\min} + n_h^{\min}, n_x^{\max} + n_h^{\max}]$$

Length rule

For sequences of lengths L_x and L_h starting at known indices, full convolution has length $L_x + L_h - 1$.

- This predicts where the output can be nonzero before any arithmetic.
- It also helps debug off-by-one errors in code.